

How to Design a High Performance GEMM Kernel (on B200)?

A practical optimization path from first principles to MMComposer

June 17, 2026

Talk Outline

- GEMM breakdown.
- The core idea: a GEMM kernel is an orchestration problem.
- From diagram to code: how the synchronization actually looks.
- Performance: how this design measures up to cuBLAS, with auto-tuning.

High-Level Design

A GEMM kernel has two major phases: **compute** and **epilogue**.

Compute

- Fetch A/B tiles.
- Stage tiles in shared memory.
- Issue MMA instructions.
- Accumulate results in TMEM.

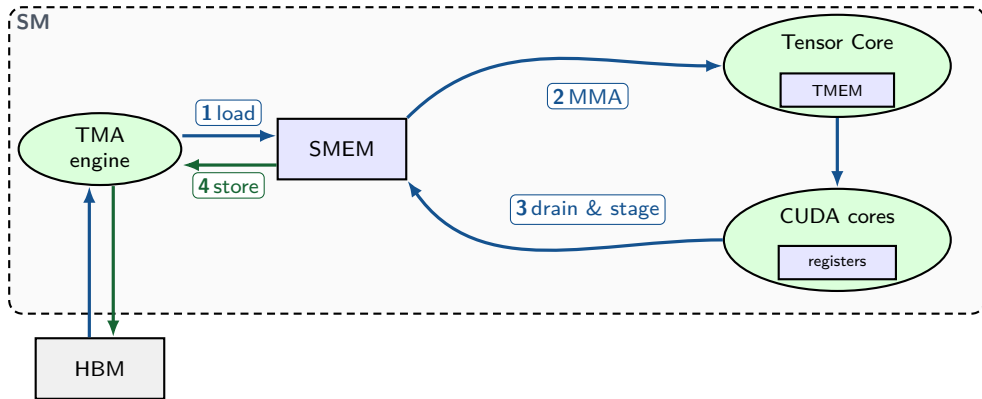
Epilogue

- Load TMEM into registers.
- Apply optional operations such as ReLU or SwiGLU.
- Pack registers into shared memory.
- Store the final result to HBM.

The Big Picture: GEMM Dataflow on the SM

Where data lives as it flows through one SM:

HBM → **SMEM** → **TMEM** → **registers** → **SMEM** → **HBM**.



Engines own their storage — **TMEM** in the tensor core, **registers** in the CUDA cores; **SMEM** is the shared hub.

The Core Idea: Orchestration

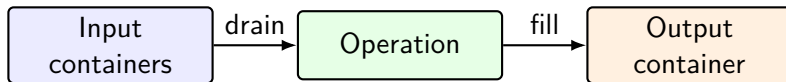
If there is one idea to take away from this talk, it is this:

A high-performance GEMM kernel is an orchestration problem.

The kernel is a dataflow schedule over hardware units, warp roles, and on-chip containers. Performance depends on keeping values flowing while reusing scarce containers at the right time.

Operations and Containers

We model the kernel as **operations** connected by **containers** that hold values:



- An operation **drains** input values from its input containers.
- It **fills** the result into one output container.

When Can an Operation Start?

An operation is schedulable only when both conditions are true:

- ① **Input value-ready**: every input value it needs has been produced.
- ② **Output resource-free**: the container for its output is available.

In other words, the operation is blocked if either the data is not ready or the destination container is still occupied.

This is the basic rule behind SMEM ring-buffer reuse, TMEM reuse, and TMA store-buffer pipelining.

Two Signals

In this model, synchronization has two meanings:

- ① **Data-ready**: a container is full; data is ready to be consumed.
- ② **Resource-free**: a container is empty; it can be reused.

A producer is waiting for a free output container. A consumer is waiting for a ready input value. Good GEMM scheduling is mostly about making those two waits overlap with useful work.

For TMEM, value readiness is gated by a completion condition: the **TMEM data ready** signal is sent only after the final MMA finishes.

—→ data/value flow

---→ synchronization signal

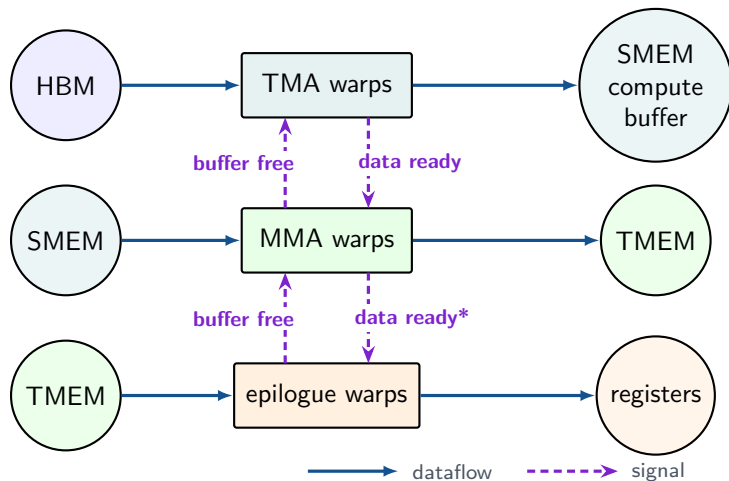
Add the Containers

For the synchronization model used in this talk, we group the kernel into three producer/consumer operations:

Operation	Input container	Output container
TMA warps	HBM A/B tiles	SMEM compute ring buffer
MMA warps	SMEM compute ring buffer	TMEM accumulator
Epilogue warps	TMEM accumulator	registers

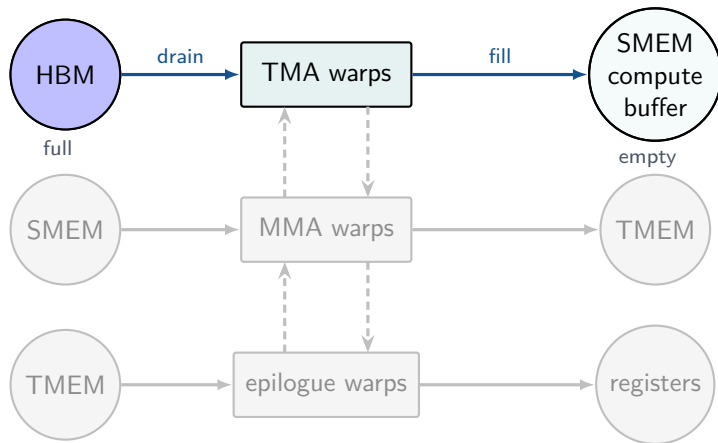
This is the abstraction used in the next graph. The epilogue's internal pack and TMA-store staging are real, but they are not the main warp-to-warp synchronization edges shown here.

Main Synchronization Pattern: Graph View



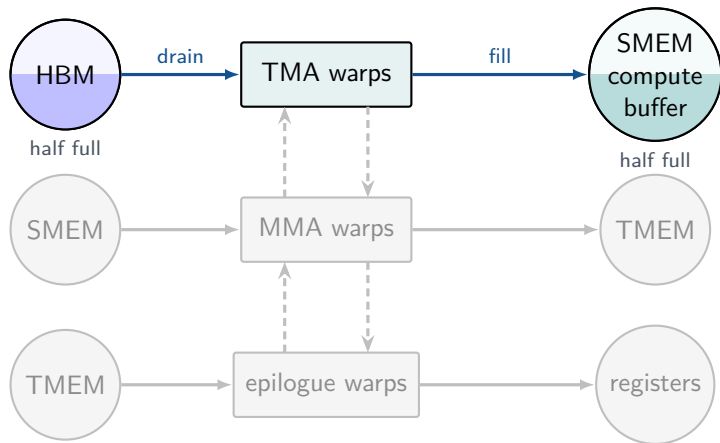
* TMEM data ready is sent only after the final MMA finishes.

TMA Warps: Fill SMEM, Signal Data Ready



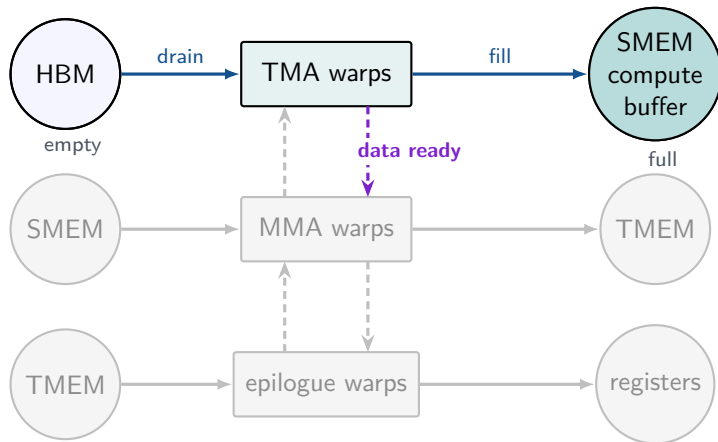
The TMA operation can start: the HBM tile is ready, and the SMEM compute slot is free.

TMA Warps: Fill SMEM, Signal Data Ready



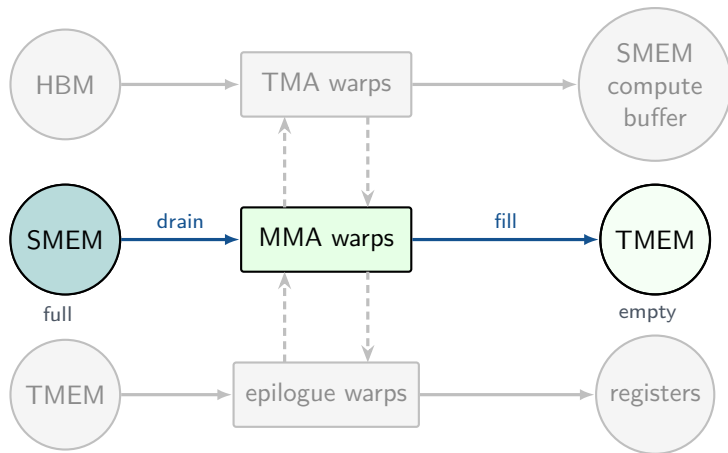
The TMA warps drain the HBM tile while filling the SMEM compute slot.

TMA Warps: Fill SMEM, Signal Data Ready



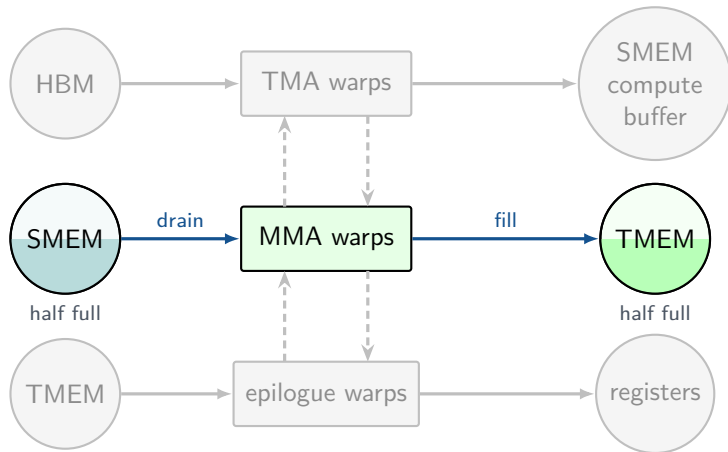
Once the HBM tile is drained and the SMEM slot is full, the TMA warps send **compute data ready** to the MMA warps.

MMA Warps: Signal Buffer Free and Data Ready



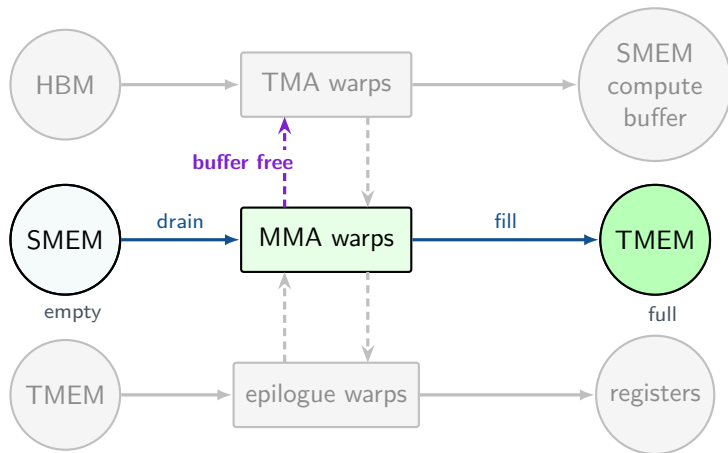
The MMA operation can start: compute data is ready in SMEM, and the TMEM accumulator buffer is free.

MMA Warps: Signal Buffer Free and Data Ready



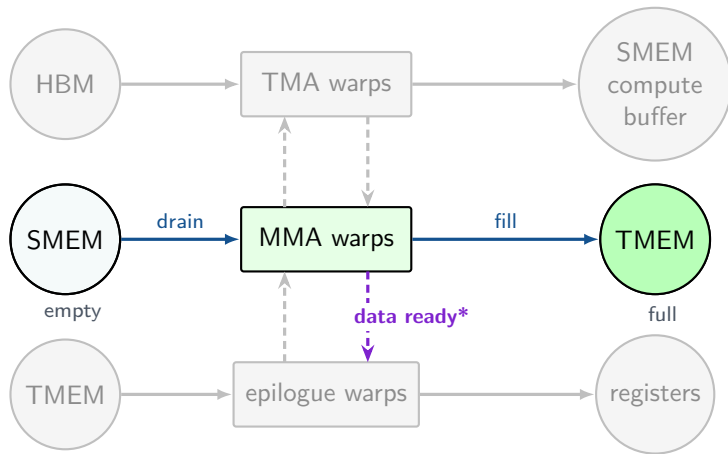
The MMA warps drain the SMEM compute buffer while filling TMEM.

MMA Warps: Signal Buffer Free and Data Ready



Once the SMEM input is drained, the MMA warps send **compute buffer free** back to TMA.

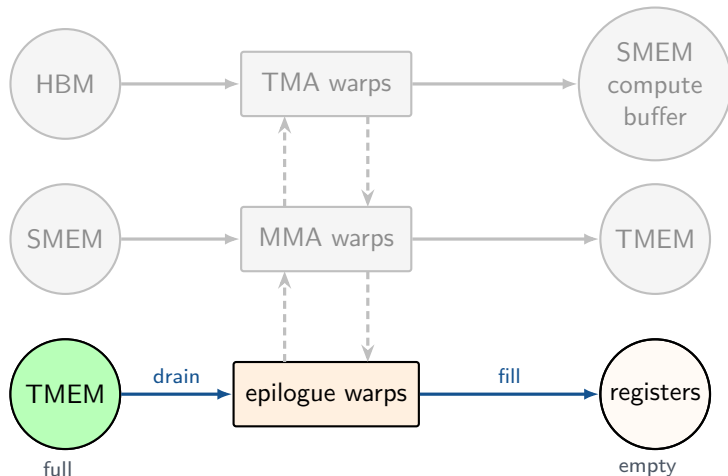
MMA Warps: Signal Buffer Free and Data Ready



After the final MMA for this output tile, the MMA warps send **TMEM data ready** to the epilogue warps.

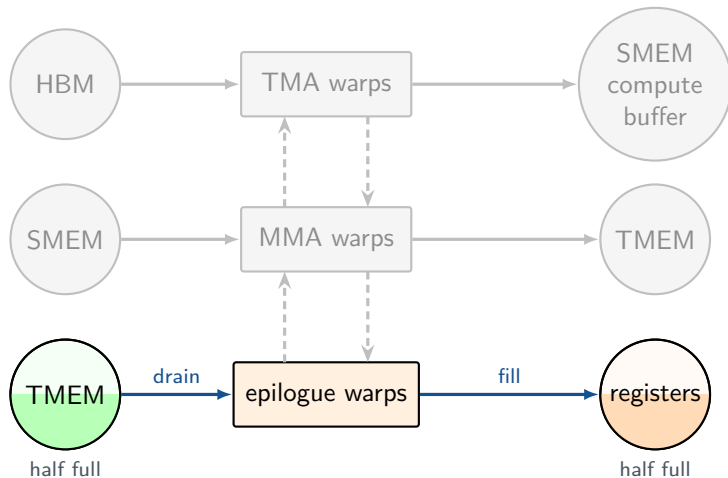
* data ready is sent only after the final MMA finishes.

Epilogue Warps: Drain TMEM, Signal Buffer Free



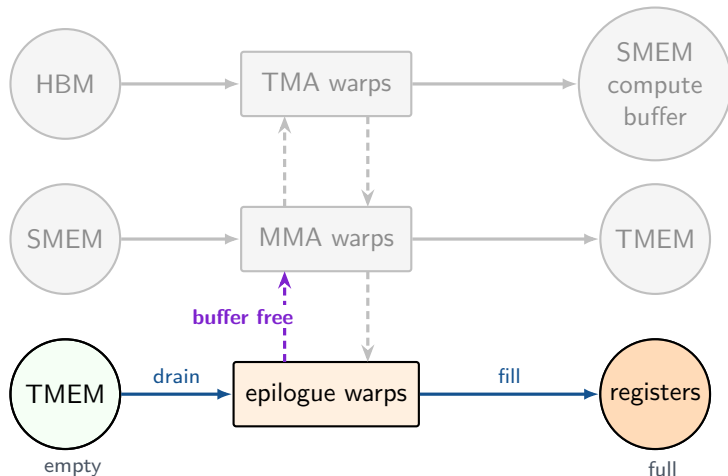
The epilogue can start: TMEM data is ready, and the register destination is available.

Epilogue Warps: Drain TMEM, Signal Buffer Free



The epilogue warps drain TMEM while filling registers.

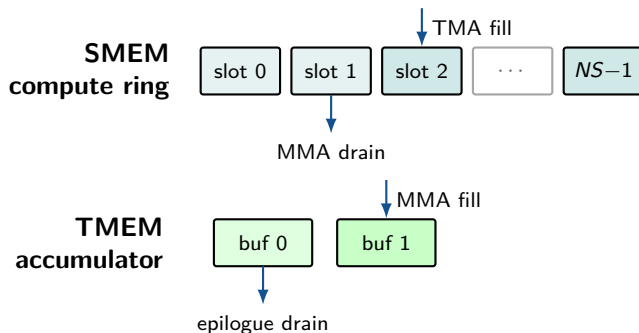
Epilogue Warps: Drain TMEM, Signal Buffer Free



Once TMEM is drained, the epilogue warps send **TMEM buffer free** back to the MMA warps.

Using Multiple Buffers

The single-container graph is the dependency for one slot. Real kernels replicate the same ready/free protocol across multiple slots.



Container	Buffering	Effect
SMEM compute ring	NS slots	MMA consumes one slot while TMA fills the next.
TMEM accumulator	2 buffers	Epilogue drains the previous output while MMA produces the next.

From Diagram to Code

We now connect the container diagram to the real kernel. Each warp role maps to one loop that *waits*, *works*, then *signals*:

Warp	Waits on	Works	Signals
TMA	smem_compute_free_mbar[slot]	TMA load A/B	smem_compute_full_mbar[slot]
MMA	smem_compute_full_mbar[slot]	tcgen05.mma	smem_compute_free_mbar, tmem_full
Epilogue	tmem_full[buf]	drain + TMA store	tmem_free_mbar[buf]

The mbarriers *are* the dashed signal arrows from the diagrams.

TMA Warp: Fill SMEM, Signal Compute Data Ready

```
if (warp_id == 0 && elect_sync()) {           // TMA-load warp
    for (int k = 0; k < num_k; k++) {
        int slot = gk % NS;
        // wait: compute buffer free
        mbarrier_wait_phase(smem_compute_free_mbar[slot],
                             smem_compute_free_phase[slot]);
        tma_2d_load_g2(A_base(slot), A_tmap, ...);    // drain HBM -> fill A
        tma_2d_load_g2(B_base(slot), B_tmap, ...);    //                fill B
        // signal: compute data ready
        mbarrier_arrive_expect_tx(smem_compute_full_mbar[slot],
                                   SLOT_BYTES);

        gk++;
    }
}
```

Wait for the output slot to be free, fill it, then signal that the compute data is ready to consume.

MMA Warp: Compute Buffer Free, TMEM Data Ready

```
else if (cta_rank == 0 && warp_id == 1 && elect_sync()) {    // MMA warp
    mbarrier_wait_phase(tmem_free_mbar[buf],
                        tmem_free_phase[buf]);                // wait: TMEM buffer free
    for (int k = 0; k < num_k; k++) {
        int slot = gk % NS;
        // wait: compute data ready
        mbarrier_wait_phase(smem_compute_full_mbar[slot],
                            smem_compute_full_phase[slot]);
        issue_mma_chain(d_tmем, A_base(slot), B_base(slot), ...);
        // signal: compute buffer free
        tcgen05_commit_mcast_g2(smem_compute_free_mbar[slot],
                                cta_mask);

        gk++;
    }
    tcgen05_commit_mcast_g2(tmем_full[buf], cta_mask); // signal: TMEM data ready
}
```

`tmем_full` is the concrete mbarrier for *TMEM data ready*; it fires only after all MMAs for this tile are done.

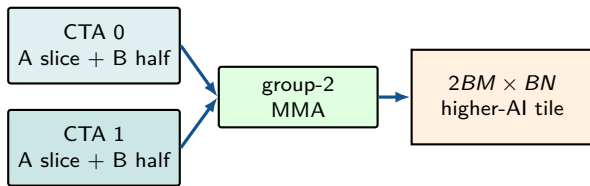
Epilogue Warps: Drain TMEM, Signal TMEM Free

```
else if (warp_id >= 4 && warp_id < NUM_WARPS + 4) {    // epilogue warps
    mbarrier_wait_phase(tmem_full[buf], full[buf]);    // wait: TMEM data ready
    tcgen05_ld(...);                                  // drain TMEM -> registers
    // ... epilogue: write the result out ...
    EPI_TMEM_FREE_ARRIVE(buf);                          // signal: TMEM buffer free
}
```

At this point the epilogue is just the TMEM *consumer*; how it writes the result out is a design choice we cover later.

One Key Optimization: 2-CTA MMA

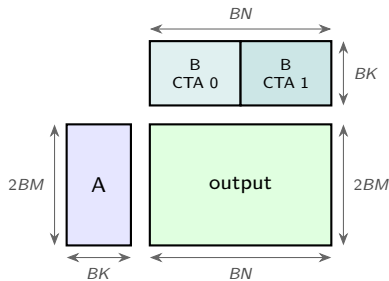
- Two CTAs form one cluster.
- Each CTA keeps one A slice and half of B in SMEM.
- One group-2 MMA produces a larger, higher-AI output tile.



Benefit: **higher arithmetic intensity**. The cluster computes a larger tile while B storage is split across the CTAs.

2-CTA MMA: Geometry

- Each CTA loads its $BM \times BK$ slice of A.
- Each CTA loads a $BK \times BN/2$ slice of B.
- At $BN=256$, per-CTA SMEM per slot drops from **48 KB** to **32 KB**.
- TMA loads use `.cta_group::2`; the `tcgen05.commit` multicasts the completion signal to both CTAs.



2-CTA MMA: In the Instructions

The cluster cooperation shows up directly in the PTX:

```
// TMA load into clustered SMEM (both CTAs' arrivals counted):
cp.async.bulk.tensor.2d.shared::cluster.global
    .mbarrier::complete_tx::bytes.cta_group::2    [smem], [tmap,{x,y}], [mbar];

// One MMA issued across the CTA pair:
tcgen05.mma.cta_group::2.kind::f16    ... ;

// Completion multicast to both CTAs in the cluster:
tcgen05.commit.cta_group::2.mbarrier::arrive::one
    .shared::cluster.multicast::cluster.b64    [smem_compute_free_mbar], cta_mask;
```

Source: [webui/kernels/tier3_cluster_swizzle/kernel.cu](https://github.com/NVIDIA/cuda-10.2/blob/master/source/runtime/ptx/kernels/tier3_cluster_swizzle/kernel.cu).

2-CTA MMA: More Compute Stages

Each CTA stores only **half** of B ($BN_{\text{local}} = BN/2$), so the compute slot shrinks and **more pipeline stages fit** in the same SMEM.

Single-CTA (full B per CTA)

BN	B tile	slot	stages
256	32 KB	48 KB	4
512	64 KB	80 KB	2

2-CTA (half B per CTA)

BN	B tile	slot	stages
256	16 KB	32 KB	7
512	32 KB	48 KB	4

A tile is 16 KB throughout. With the SMEM capacity (≈ 227 KB), $\text{stages} = \lfloor \text{SMEM} / \text{slot} \rfloor$. 2-CTA fits $\sim 2\times$ the stages.

Tradeoff of Double TMEM Buffering

Tile width trades off **overlap** against **arithmetic intensity**.

$$\text{2-CTA MMA, } BM=128 : \quad AI = \frac{256BN}{256 + BN}$$

$BN=256$

MORE OVERLAP

double TMEM buffer
up to 7 compute stages
lower AI: 128 flop/B

$BN=512$

HIGHER AI

single TMEM buffer
up to 4 compute stages
delayed reuse: 171 flop/B

For medium-to-large square GEMMs, higher AI often wins.

Epilogue Design: Three Phases

So far we have shown only the first phase of the epilogue. In full it has **three**:

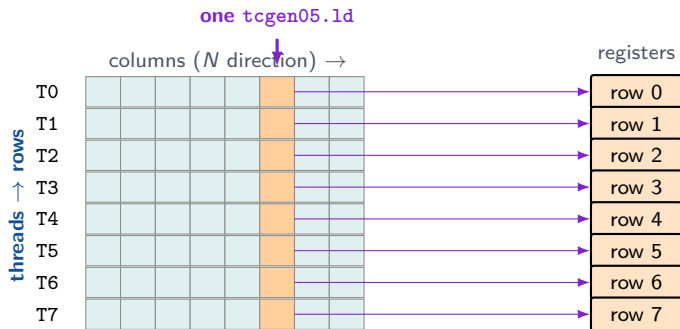


- ① **Drain TMEM into registers** with `tcgen05.ld` (already seen).
- ② **Write registers into an SMEM staging buffer.**
- ③ **Issue a TMA store** from SMEM to HBM.

Phases 2 and 3 are forced on us by *how TMEM is laid out*.

Epilogue Design: TMEM Layout

TMEM is laid out for MMA accumulation: `tcgen05.1d` reads it **one column at a time**, with each **thread mapped to a row**.

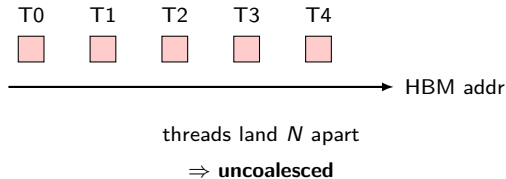


A load distributes one column across the warp's lanes; iterating, each **thread t ends up owning row t** — not a contiguous run of neighboring threads' outputs.

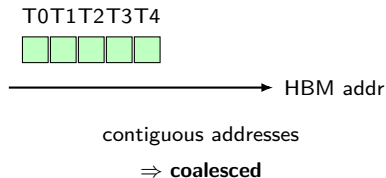
Epilogue Concern 1: Coalesced Global Write

Each thread owns a *row*, so storing registers straight to HBM is row-strided. Instead we first fill one $BM \times 64$ **chunk** in SMEM — the TMA store unit — then let **TMA** write it out, coalesced.

Direct store



Via SMEM staging



Epilogue Concern 2: Filling the 128×64 SMEM Buffer

We fill it `tcgen05.1d.x8` chunk by chunk. One warp store writes **one int4 column** (one cell per lane — the highlighted column, not the whole row); naively those cells share **one bank**, so the swizzle $n \oplus (\text{row} \& 7)$ spreads them.

Naive: column $\rightarrow$ fixed bank

T0	0	1	2	3	4	5	6	7
T1	0	1	2	3	4	5	6	7
T2	0	1	2	3	4	5	6	7
T3	0	1	2	3	4	5	6	7
T4	0	1	2	3	4	5	6	7
T5	0	1	2	3	4	5	6	7
T6	0	1	2	3	4	5	6	7
T7	0	1	2	3	4	5	6	7

a column hits **one bank**

$\Rightarrow$ 8-way conflict

Cell = $c \oplus r$ = bank group; each is one int4 = **4 banks** (8 cells = 32 banks) = the SWIZZLE_128B atom.

Shown is one phase (lanes 0–7); A warp writes 32 rows and a 128-bit store issues as 4 identical phases.

Swizzled: $n \oplus (\text{row} \& 7)$

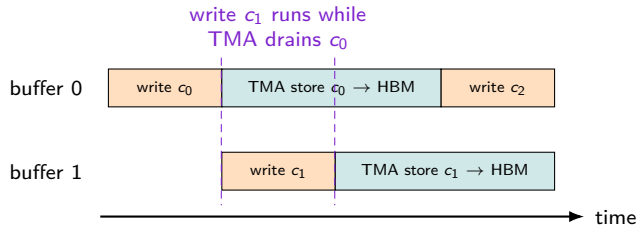
T0	0	1	2	3	4	5	6	7
T1	1	0	3	2	5	4	7	6
T2	2	3	0	1	6	7	4	5
T3	3	2	1	0	7	6	5	4
T4	4	5	6	7	0	1	2	3
T5	5	4	7	6	1	0	3	2
T6	6	7	4	5	2	3	0	1
T7	7	6	5	4	3	2	1	0

each row $\rightarrow$ a **different bank**

$\Rightarrow$ conflict-free

Epilogue Design: Multiple Store Buffers

With ≥ 2 store buffers, one chunk's **TMA store** overlaps the next chunk's **SMEM write**:



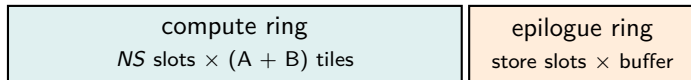
- The TMA store is **async**: once issued, the epilogue moves to the next chunk's write.
- With N buffers ($N \in \{1, 2, 3, 4\}$) it fills a free one while TMA drains earlier ones — stalling only when all N are in flight.

Epilogue Design: From Design to Code

The whole epilogue is one loop over the $BN/64$ store chunks:

```
for (int chunk = 0; chunk < BN/64; chunk++) {  
    // Phase 1: load all 64 cols of this chunk into registers  
    for (int n = 0; n < 8; n++)           // 8 x tcgen05.ld.x8  
        tcgen05_ld_x8(tmem_row + chunk*64 + n*8, reg[n]);  
    tcgen05_wait_ld();  
    if (elect) tma_wait_group<TMA_STORE_STAGES-1>(); // a buffer is free  
    bar.sync;                                     // ... visible to all warps  
    // Phase 2: conflict-free SMEM write, then the TMA store  
    for (int n = 0; n < 8; n++) {  
        int swz = n ^ (row & 7);           // XOR swizzle: distinct banks  
        store_buf[stage][row][swz] = pack_bf16(reg[n]);  
    }  
    bar.sync;                                     // buffer fully written  
    if (elect) tma_store(store_buf[stage], C + chunk*64);  
    stage = (stage + 1) % NUM_STORE_BUFFERS;  
}
```

Partitioning Shared Memory



- **Compute ring:** NS pipeline slots, each holding one A tile and one B tile for the MMA stage.
- **Epilogue ring:** store slots, each a TMA store buffer holding one $BM \times 64$ bf16 chunk: $128 \cdot 64 \cdot 2 = 16 \text{ KB}$.
- Both rings share one per-SM SMEM budget, so larger tiles or deeper pipelines on one side leave less room for the other.

Tile Sizes and the SMEM Budget

Per compute slot (2-CTA, each CTA owns half of B):

$$\text{slot} = \underbrace{BM \cdot BK \cdot 2}_{\text{A tile}} + \underbrace{\frac{BN}{2} \cdot BK \cdot 2}_{\text{B tile (local)}}$$

Config (2-CTA, $BK=64$)	A tile	B tile	slot
$BM=128, BN=256$	16 KB	16 KB	32 KB
$BM=128, BN=512$	16 KB	32 KB	48 KB

Budget constraint (per SM, B200 $\approx$ 227 KB):

$$\underbrace{NS \times \text{slot}}_{\text{compute ring}} + \underbrace{E \times 16 \text{ KB}}_{\text{epilogue ring}} \leq S_{\text{SMEM}}$$

Example: $BN=256, NS=5 \Rightarrow$ 160 KB compute ring; with $E=2$ store buffers, +32 KB = 192 KB total.

Common Partitions for $BM=128$ (empirical)

Each split below is the **maximum** that fits in ≈ 227 KB; the config actually used is chosen by autotune per shape:

$BN=256$  up to $6 \times 32 + 2 \times 16$ KB

$BN=256$  up to $5 \times 32 + 4 \times 16$ KB

$BN=512$  up to $4 \times 48 + 2 \times 16$ KB

 compute slot ($BN=256$: 32 KB, $BN=512$: 48 KB)  16 KB store buffer — fewer/larger compute slots trade for more store buffers.

Autotuning: Configuration Search Space

Autotune sweeps these parameters and benchmarks every *valid* combo:

Parameter (covered in this talk)	Candidate values
BM (tile rows)	{128}
BN (tile cols)	{256, 512}
compute slots (NS)	{2, 3, 4, 5, 6}
epilogue slots (store buffers)	{1, 2, 3, 4}

- A **static checker** prunes combos whose SMEM exceeds **227 KB** ($NS \times \text{slot}(BN) + \text{store} \times 16 \leq 227$) before benchmarking; the fastest survivor wins.
- The framework fixes everything else; the one knob it leaves open is the **CTA-swizzle factor** {1, 2, 4, 8, 16} (an L2-reuse block ordering, not covered here).

Implementation and Performance Results

Warp layout — deliberately not a power of two. Each CTA launches

$$\underbrace{1}_{\text{TMA}} + \underbrace{1}_{\text{MMA}} + \underbrace{4 \text{ or } 8}_{\text{epilogue}} = 6 \text{ or } 10 \text{ warps.}$$

- The **TMA** and **MMA** warps are fixed — one each.
- The **epilogue** warp count $\in \{4, 8\}$ is **autotuned**; more warps give the TMEM drain and SMEM write more parallelism.
- So the CTA is sized to the work, not rounded up to a power of two.

Performance: Square GEMMs vs cuBLAS

MMComposer averages **~100% of cuBLAS** (96–106% across 20 shapes). Each top config *is* an SMEM partition.

shape	cuBLAS	MMC	%	<i>BN</i>	<i>NS</i>	store
4096 ³	1382	1354	98	256	6	1
5120 ³	1432	1373	96	512	4	2
6144 ³	1475	1447	98	512	4	2
7168 ³	1435	1397	97	256	5	1
8192 ³	1420	1453	102	512	4	2
9216 ³	1422	1426	100	512	4	2
10240 ³	1398	1390	99	512	3	1
11264 ³	1414	1440	102	512	3	2
12288 ³	1477	1472	100	512	4	2

TFLOPS (B200, BF16). *BN* / *NS* / **store** is the SMEM partition: *NS* compute slots of a *BN*-wide tile + *store* TMA buffers.

Performance: Rectangular GEMMs vs cuBLAS

shape ($M \times N \times K$)	cuBLAS	MMC	%	BN	NS	store
$2048 \times 36864 \times 12288$	1517	1480	98	512	4	2
$2048 \times 49152 \times 12288$	1366	1390	102	512	4	1
$4096 \times 12288 \times 4096$	1486	1458	98	256	6	1
$4096 \times 16384 \times 4096$	1512	1529	101	256	5	2
$8192 \times 24576 \times 8192$	1490	1507	101	512	4	2
$8192 \times 32768 \times 8192$	1404	1458	104	512	3	2
$16384 \times 4608 \times 768$	1467	1432	98	256	5	1
$32768 \times 4608 \times 768$	1461	1423	97	256	5	1
$32768 \times 9216 \times 768$	1428	1415	99	256	6	1
$32768 \times 12288 \times 768$	1321	1401	106	256	6	1
$65536 \times 2304 \times 768$	1460	1424	98	256	5	1

Wide/short LLM-style shapes. BN / NS / **store** is the same SMEM partition as the square case — autotuned per shape.

Why More Buffers Doesn't Always Help

The autotuned configs rarely max out the buffer counts (store mostly 1–2, not 4) — something we would not have predicted before seeing the numbers. Some intuition (*hypotheses*, not proven):

Compute ring (more *NS*)

- Once the pipeline hides load/MMA latency, extra stages add nothing.

Epilogue ring (more store buffers)

- If the epilogue is already fast relative to compute, the extra store $\leftrightarrow$ write overlap buys little.
- More TMA stores in flight = more memory parallelism that may contend with the TMA *loads* feeding the compute pipeline.

The optimum is shape-dependent and empirical — hence autotune.

Adding Epilogue Fusions

Adding Epilogue Fusions

A custom epilogue fusion is almost entirely an **epilogue** change — the compute pipeline is reused, with at most a minor TMA-load tweak.

Case study: fused SwiGLU. One GEMM produces two projections side by side in TMEM (*left* and *gate*); the epilogue writes out both and the gated result:

$$\text{SwiGLU} = \text{left} \odot \text{SiLU}(\text{gate}), \quad \text{SiLU}(g) = g \sigma(g).$$

- Loop over *pairs* of column chunks (a left chunk + its gate chunk).
- Per pair: load both into registers, then **three** stores — left, gate, and SwiGLU.
- *Dual-B*: left and gate are *separate* weight matrices; with 2-CTA on, each CTA loads its half from a different one — a tiny load change, with **no measured performance impact**.

Fused SwiGLU: the Epilogue Loop

Only the epilogue changes; `stage_and_store` is the canonical swizzled-write + TMA-store helper:

```
for (int p = 0; p < (BN/64)/2; p++) {           // loop over chunk PAIRS
  for (int n = 0; n < 8; n++) {                 // load both projections
    tcgen05_ld_x8(tmem + p*64 + n*8, left[n]);
    tcgen05_ld_x8(tmem + (p+HALF)*64 + n*8, gate[n]);
  }
  tcgen05_wait_ld();

  stage_and_store(left, C, p*64);               // (1) store left
  stage_and_store(gate, C, (p+HALF)*64);        // (2) store gate

  for (int n = 0; n < 8; n++)                  // (3) fused SwiGLU math
    swig[n] = left[n] * silu_fast(gate[n]);     //   ex2.approx + rcp.approx
  stage_and_store(swig, D, p*64);              //   store SwiGLU -> D
}
```

`stage_and_store` = the same store step as the canonical epilogue; fusion just adds the math and an extra output tensor D.

Fused SwiGLU: Almost Free

$M=32768$, $N=4608$, $K=768$; $BM=128$, $BN=256$, $NS=6$, store=2

	TFLOPS	time
cuBLAS (GEMM only)	1406	$165\ \mu s$
fused SwiGLU ($1.5\times$ store)	1250	$185\ \mu s$

- Despite $1.5\times$ **the store traffic** (left + gate + SwiGLU) and the SwiGLU math, it is only **$\sim 12\%$ slower** — 89% of a pure cuBLAS GEMM.
- The SwiGLU math is essentially **free**; the gap is almost entirely the **extra store**.